

# Abid Karim

Full Stack Developer (Node.js + Python) — SQL — RAG Systems — AI Agents

Fresh graduate CS student from LUMS with production experience in RAG systems and full-stack TypeScript apps, having mentored **550+** students and led teams across 2 industrial standard projects — seeking backend or AI engineering roles.

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## EXPERIENCE

### 10Pearls

Lahore, Pakistan

Software Engineer Intern

Sep 2025 – Nov 2025

- Built a real-time full-stack TypeScript MERN notes application with advanced filtering and rich-text editing features
- Implemented secure JWT authentication via HTTP-only cookies and WebSocket-based real-time notifications
- Designed OpenAPI-compliant RESTful APIs, containerized services using Docker, and integrated SonarQube in CI/CD pipelines, reducing vulnerabilities by **25%**

### LUMS

Lahore, Pakistan

Senior Year Software Engineering Project – Team Lead

Sep 2025 – April 2026

- Led a **5-member** team to build an AI-powered maternal health platform using FastAPI, React, and PostgreSQL
- Designed a **multi-tenant RAG system** using Google Gemini API and ChromaDB, and built a database AI agent to handle natural language maternal health queries
- Engineered a data architecture using PostgreSQL and ChromaDB, and implemented **CI/CD pipelines** with Docker and Kubernetes for scalable deployment

### LUMS

Lahore, Pakistan

Teaching Assistant – Software Engineering

Jan 2026 – April 2026

- Mentored **300+** students across multiple semesters in Java and Android development; conducted code reviews for **50+** projects and managed Git/GitHub workflows
- Improved project completion rates by **35%** through targeted workshops and office hours
- Developed automated grading scripts that reduced assessment time by **40%** while maintaining evaluation quality

### Wetual – NIC Lahore

Lahore, Pakistan (Hybrid)

Business Analyst Intern

Jun 2023 – Aug 2023

- Built Python and SQL pipelines to analyse financial datasets, surfacing insights that informed early product and pricing decisions
- Supported drafting of **SAFE agreements**, contributing to legal and financial documentation for fundraising processes
- Surveyed **50+** startups to gather market insights, achieving a **95%** positive response rate

## PROJECTS

### CampusCaravan – Carpooling Platform | Express, React, PostgreSQL, WebSockets

Jan 2024 – May 2024

- Led team of **5** to develop carpooling platform adopted by **150+** students, reducing commuting costs by **40%**; built backend with Node.js, TypeScript, and PostgreSQL
- Implemented real-time chat via WebSockets, AI content moderation with Gemini API, and JWT authentication with HTTP-only cookies
- Optimized database queries and search algorithms, improving search performance by **30%** and reducing average API response time by **40%**

### LUMSPiRE – Visual Discovery Platform | MERN Stack, TypeScript, WebSockets

Jun 2025 – Jul 2025

- Developed a Pinterest-style visual discovery platform with personalized feeds, serving **100+** users with an average page load time under **1.5s**
- Integrated WebSockets for real-time messaging and notifications, implemented JWT authentication with HTTP-only cookies
- Designed responsive UI with optimized image loading and caching, reducing asset load time by **35%**

## SKILLS

**Languages:** TypeScript, Python, SQL, C++, Java, HTML/CSS

**Frameworks & Tools:** Node.js, Express.js, FastAPI, React.js, PostgreSQL, Docker, Git, Postman, Fastify.js, Django, Kubernetes, CI/CD, SonarQube, Jira

**AI Stack:** RAG Systems, AI Agents, Google Gemini API, ChromaDB, Pandas, LangChain

## EDUCATION

### Lahore University of Management Sciences (LUMS)

Lahore, Pakistan

BSc Computer Science and Finance

Sep 2022 – May 2026

- Relevant Courses:** Software Engineering, Data Structures (C++), Advanced Programming, Blockchain, Databases, Computer Systems, Discrete Mathematics, Probability & Statistics, Linear Algebra, Calculus